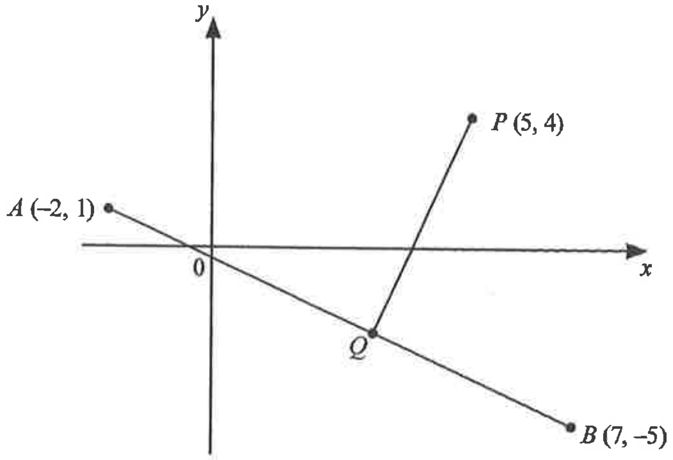
Unit 2.6 Coordinate Geometry

Paper 1: Intermediate

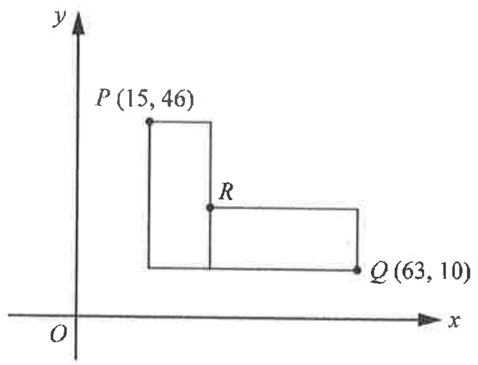
1.



The diagram shows the points $A(-2, 1)$, $B(7, -5)$ and $P(5, 4)$.
Q is a point on AB so that PQ is perpendicular to AB.
The product (gradient of PQ) × (gradient of AB) = –1.
Use this information to find the equation of the line PQ.

[4]
[N19/I/22]

2. The diagram shows two congruent rectangles.
The sides are horizontal and vertical.
Point P has coordinates (15, 46) and Q has coordinates (63, 10).

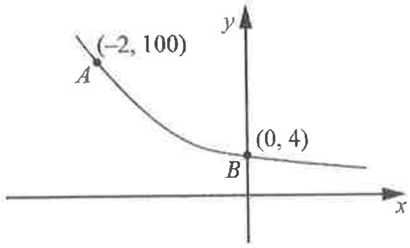


Find the coordinates of R.

[2]
[N16/I/14]

3. The sketch shows the graph of $y = ka^{-x}$.

The points $A(-2, 100)$ and $B(0, 4)$ lie on the graph.



(a) Find the values of k and a. [2]

(b) A straight line is drawn from A to B.

Find the equation of the line AB. [2]

[N16/I/17]

Paper 2: Intermediate

1. A is the point $(-4, 10)$ and B is the point $(4, -2)$.

(a) Find the length of the line AB. [2]

(b) Find the equation of the line AB. [2]

(c) The equation of line p is $3x + 2y = 5$.

(i) Show how you can tell that the line p does **not** intersect the line AB. [2]

(ii) The equation of line q is $6y = 4x - 37$.

Find the coordinates of the point of intersection of the line p and the line q. [3]

[N17/II/4]

Paper 2: Advanced

2. The coordinates of point A are $(5, 5)$.

Line p passes through point A and has gradient $-\frac{2}{9}$.

(a) Show that the equation of line p is $9y + 2x = 55$. [2]

(b) The equation of line q is $3y = 4x - 26$.

Find the coordinates of the point of intersection of line p and line q. [3]

(c) Line p intersects the y-axis at point B and line q intersects the y-axis at point C.

Line q intersects the line $x = 5$ at point D.

(i) Explain why $ABCD$ is a trapezium and state the distance between the parallel sides. [2]

(ii) Calculate the area of trapezium $ABCD$. [4]

[N21/II/5]